contents

ELECTRONICS FOR YOU PLUS | DECEMBER 2012 | VOL. 1 NO. 8

20 Technology Focus
Next-Gen Displays Enabling Super-Friendly Devices



82 Buyers' Guide
Pico Projectors for People on the Move



30 LEDs
Raising the Luminous Efficacy of LEDs

38 Chip Design
Why 3D Chips are the Future of Electronics

44 Test & Measurement
The Latest in Protocol Analysers

50 FPGA
Power Measurement and Control Using Programmable Logic

54 Embedded
How to Select A Microcontroller

62 Solar
Solar Energy for Your Home

68 EFY report
Educational Kits: Innovation Lacking, Sales Thriving

88 Innovation
Prick-free Blood Tester from India

92 Do-it-Yourself
Android with Arduino

94 Telecom
Bluetooth 4.0: Low-power Feature Enables Revolutionary Uptake

98 Do-it-Yourself
Arduino-Based Fancy Lights Controller

123 Tutorial
Understanding NAND Gates with CD4011

125 IT
Accelerating USB 3.0 Implementations: Time is money

Hands-On



Construction 102
Real-Time Clock

Construction 108
Arduino-based RF Controlled Robot

Construction 111
MSP430G2231-based Temperature Indicator and Controller

Software Section 115
- JavaScript Based Man-Demon Puzzle Simulator
- Circuit Design Calculator Using 'C++'

Circuit Ideas 118
1.5W Power Amplifier,
Simple Stereo Level Indicator,
FM Bug,
Calling Bell Using an Intercom,
USB Charger for Mobile Phones

128 Career
Optoelectronics is Where the Expertise Lies!

150 Interview
"India is a really important market for a device like Raspberry Pi"
— Eben Upton, executive director, Raspberry Pi Foundation

152 Interview
"The new standard ECQ HSPM 80000 is an ideal approach to adopt system-level compliance"
—Shanmuga Sundaram, country head, Material Testing Laboratories, TUV Rheinland India

Regulars

Websites	6	New Products	130	Product Categories	
Letters	8	Industry News	134	Index	160
Q&As	10	Business Pages	139	Organisations Index	161
Technology News	14	Electronics Mart	154	Advertisers' Index	161
First Look	76				

EFY Plus DVD

Electronics Development Tools I
Audio and Video Editing Tools VI

NEXT ISSUE • Ten Scientific Findings that Could Influence the Future of Electronics • Passive Components • Variable DC Power Supplies

EDITOR : Ramesh Chopra

EDITORIAL CORRESPONDENCE : Editorial Secretary
Phone: 011- 26810601; E-mail: editsec@efyindia.com
(Technical queries: efylab@efyindia.com)

SUBSCRIPTIONS & MISSING ISSUES : Phone: 011- 26810601 or 02 or 03
E-mail: support@efyindia.com

BACK ISSUES, BOOKS, CDs, PCBs etc. : Kits'n'Spares, New Delhi
Phone: 011- 26371661, 26371662
E-mail: info@kitsnspares.com

EXCLUSIVE NEWSSTAND DISTRIBUTOR : IBH Books & Magazine Distributors Pvt Ltd, Mumbai
Phone: 022- 40497401, 40497402, 40497474, 40497413; Fax: 40497414
E-mail: circulations@ibhworld.com

ADVERTISEMENTS NEW DELHI (HEAD OFFICE) : Ph: 011- 26810601 or 02 or 03
E-mail: efyenq@efyindia.com

MUMBAI : Ph: 022- 24950047, 24928520
E-mail: efymum@efyindia.com

BENGALURU : Ph: 080- 25260394, 25260023
E-mail: efyblr@efyindia.com

CHENNAI : Ph: 0974286 4199
E-mail: efychn@efyindia.com

HYDERABAD : Ph: 08800094211
E-mail: efyenq@efyindia.com

KOLKATA : Ph: 08800094201
E-mail: efycal@efyindia.com

PUNE : Ph: 022- 24950047, 24928520
E-mail: efypune@efyindia.com

GUJARAT: Ph: 022- 24950047, 24928520
E-mail: efyahd@efyindia.com

CHINA : Power Pioneer Group Inc.
Ph: (86 755) 83729797, (86) 13923802595
E-mail: terryqin@powerpioneergroup.com

JAPAN : Tandem Inc., Ph: 81-3-3541-4166

SINGAPORE : Publicitas Singapore Pte Ltd
Ph: +65-6836 2272
E-mail: singapore@publicitas.com

TAIWAN : J.K. Media, Ph: 886-2-87726780 ext. 10

UNITED STATES : E & Tech Media
Ph: +1 860 536 6677
E-mail: veroniquelamarque@gmail.com

Printed, published and owned by Ramesh Chopra. Printed at Nutech Photolithographers, D-58, Okhla Industrial Area, Phase-I, New Delhi, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase I, New Delhi 110020. Copyright 2012. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publishers is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.